Experiment Designs and Pre-Registration

Experiment S (scaffold characterisation) and Experiment P (pooled recovery)

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Abstract

This document is a **pre-registration**. It states the design, the metrics, and the predicted outcomes of Experiments S and P *before either is run*, so that the results can be reported as hits and misses rather than as a narrative assembled afterwards. The project has twice benefited from doing this: two of the four future-work propositions were falsified and the repairs became the backbone of the method, and the U-shape prediction for orientation error was falsified on 13 August in a way that inverted a design axis.

Both experiments previously existed only as prose in §8 and §9 of the 31 July detection-phase working document. This note extracts them, updates them for what has been measured since, and records what would count as a failure.

Provenance tags.

[VALIDATED]	measured against ground truth
[SPECIFIED]	designed, not yet run
[NEW]	changed or added since the 31 July design
[OPEN]	acknowledged gap

1 Why these two, and in this order

The method note (`boundary_recovery_v5`) sets out a four-step critical path. Steps 1 and 2 are done:

- **Step 1** — the distance estimator $\hat{d} = -\sigma\Phi^{-1}(\hat{\pi})$ recovers true distance at $r = 0.996$, median relative error -0.92%, flat across a $10\times$ noise sweep. [VALIDATED]
- **Step 2** — the separability-derived normal carries *no detectable bias* in any of 20 cells; pooled orientation error falls below 3° at $N = 100$ down to $\Delta/\tau = 1.5$. [VALIDATED]

Experiment P is step 3. Experiment S is not on that path at all — it characterises the *real* target, and every power number in the project to date is against planted synthetic gates that we designed ourselves.

S runs first, and specifically S5 and S1. They are cheap, and between them they decide what the rest of the summer is for. If the scaffold's jump is 3–5σ, detection power is not a live problem and the binding constraints are coverage and the on-manifold tension. And if the OOD detector does not recognise our probe geometry, the paper's claim changes from "we detect the switch" to "we are immune to it" — which reframes everything upstream.

PART I

Experiment S — characterising the actual target

2 Premise and build

> Every power figure in this project is measured against gates we planted. Measure what the real attack induces *before* optimising further against synthetic ones.

The target is the Slack et al. (AIES 2020) scaffolding attack: a biased model f that uses a sensitive attribute, an innocuous ψ that uses an uncorrelated synthetic feature, and an OOD detector — a random forest trained to separate real data from LIME-perturbed samples — routing between them.

Audit the score, not the label. The framework presumes a continuous response. The housing scaffold in `src/attacks/scaffold.py` returns continuous predictions, so this is satisfied. If a dataset exposes only hard labels, stop and record that a discrete-response Stage 3 variant is a prerequisite rather than a tweak.

Data availability, checked 13 August. [NEW] The **housing scaffold is runnable now**: `seattle_sample_3k.csv` and the demographics join are present, the scaffold and both OOD detector constructors are in `src/attacks/`, and `ood_detector.predict_proba` is directly callable. The **COMPAS, German Credit and Communities-and-Crime datasets are absent** — `papers/` contains only `pexsbm`, not the Slack repository, and only downstream result directories survive. S therefore runs on housing first; the other three require re-obtaining the Slack repository and are deferred, not dropped.

3 The five measurements

S1 — what is the scaffold's $\Delta/\sigma_{\text{resid}}$?

Sample point pairs straddling the OOD decision boundary at matched covariate distance; measure the jump in audited output relative to the LTS residual scale at the same anchors.

This is the single number that determines whether detection power is a live problem at all. Reference points from the measured floor: the dip has *zero* power below $\approx 2\sigma$ (the mixture is literally unimodal there), and the equal-variance LRT reaches 0.42–0.54 at 1.5σ and 1.00 at 2.5σ.

S5 — what does the detector call *our* probes?

Generate the audit's probe cloud at real anchors and ask the OOD detector to classify it.

The audit's perturbation is not LIME's. LIME perturbs every feature, ambiently, with unit Gaussian noise; the detector is trained on exactly that. Ours is isotropic within a $\hat{d}$-dimensional tangent frame at small radius. **The detector was never trained against our probe geometry.**

S2, S3, S4

- **S2 — boundary accessibility.** Fraction of real held-out points lying within r_{hi} of the OOD boundary. This is coverage for the real target, as opposed to coverage in a simulation where anchors are placed by design.

- **S3 — density-filter tension.** For anchors whose ball crosses the OOD boundary, record the retention ρ.
- **S4 — detector leakage.** Fraction of real points the OOD detector misclassifies. This is the on-manifold region where crossing is possible without leaving support.

> **S3 is an explicit falsification condition.** If ρ falls below the abstention threshold at essentially every anchor whose ball crosses the OOD boundary, then `A3` fires wherever the signal is and **the method abstains on its own target**. Report that as a falsification, not as a tuning problem.

4 The pre-registered trichotomy

S5 determines which of three regimes the scaffold falls into. All three are reportable; two are wins.

	What S5 shows	What it means
(i)	probes are classified as real; we query f throughout the ball	No gate to detect — but our attributions are of the *biased* model. **The scaffold is defeated by evasion rather than by detection.** A paper result, differently framed.
(ii)	probes straddle the OOD boundary	Gate detected. The method works as designed on the real target.
(iii)	probes are classified OOD; we see only ψ	Fooled. The density filter is supposed to prevent this — verify that it does.

Prediction, stated before running. [NEW] **Regime (i).** The LIME detector is trained on ambient $N(0, 0.3)$ perturbations across *all* features; our probes live in a $\hat{d}$-dimensional tangent frame with zero component off it. Different geometry entirely. Concretely: **we predict the detector classifies** $>80\%$ **of our probes as real**, and that the audited response is f rather than ψ over the great majority of each probe ball.

The structural reading, if (i) holds. Regimes (i) and S3 are two faces of one fact: the gate fires off-manifold, so an on-manifold auditor either never sees it (immune) or must leave the manifold to see it — at which point the density filter rejects the probes and the claim degrades from *a gate fires on deployed inputs* to *a gate exists somewhere in input space*. Stating that cleanly is arguably a stronger contribution than a detection number: **scaffolding attacks hide in the region an on-manifold audit is definitionally excluded from.**

PART II

Experiment P — pooled recovery

5 Design principle

> False positives and true positives differ *geometrically*, not merely in strength. A real boundary produces a consistent normal across every anchor touching it; a spurious detection produces a random one. Pooling on geometric agreement suppresses false positives faster than it accumulates true ones — which is what is wanted when per-anchor power is around 0.5.

Two corollaries. **Do not threshold per anchor before pooling**: at power 0.5 that discards half the true signal. And the figure of merit becomes the number of anchors required to recover the boundary, not per-anchor power.

6 The per-anchor primitive

Computed for **every** anchor, including those that do not fire.

```
1. 2-GMM EM on the LTS residuals            -> responsibilities gamma
2. linear discriminant on (z, gamma)        -> normal nu, offset b
   *** fit on ALL probes, not the LTS inlier set ***          [NEW]
3. lift to ambient:  n_a = U nu / ||U nu||
   oriented so the higher-mean component is on the + side
4. signed offset:    c_a = n_a . x_a + b / ||nu||
5. weight:           w_a = LRT statistic, NOT thresholded

   each anchor emits (n_a, c_a, d_hat_a, Delta_hat_a, w_a)
```

Steps 1–3 are [VALIDATED] as of 13 August. Step 2's "all probes" qualifier is [NEW]: median orientation error is 5.8° fitting on every probe against 21.9° on the LTS inlier set, because the trim removes points that are predominantly the minority branch — exactly the branch carrying the crossing signal. Soft responsibilities also beat hard $\gamma > 0.5$ labels (5.8° against 7.6°).

7 Estimators and null

	Estimator	Uses geometry
A	Fisher/Stouffer combination of per-anchor p-values	no — baseline
B	threshold on w_a, then cluster (n, c) in projective space (antipodal identification), requiring $\lvert\cos(n_a, n_b)\rvert > 1 - \epsilon$ and $\lvert c_a - c_b\rvert < \delta$	partly
C	w-weighted mode-seeking (mean-shift or Hough voting) over *all* anchors, no thresholding	fully

Null. Permute the residual-to-point assignment within each anchor and rerun the pipeline from step 1. This randomises the normals while preserving the marginal residual distributions. ≥ 500 permutations.

8 Generative setting

[NEW]

Flat intrinsic $d = 2$ manifold in ambient $D = 20$; a single planted hyperplane; the tangent frame supplied, per the Stage-0 deferral.

Axis	31 July design	Revised, 13 August
$\Delta/\sigma_{\text{resid}}$	$\{1.0, 1.5, 2.5, 5.0\}$	unchanged
π	$\{0.10, 0.25, 0.50\}$	$\mathbf{\{0.05,\ 0.10,\ 0.20\}}$
N anchors	$\{5, 10, 25, 50, 100, 200\}$	unchanged
anchor distance / r_{hi}	$\{0.2, 0.5, 0.9, 1.5\}$	**shell $d/\sigma \in [1.28, 1.64]$**, plus a non-crossing control

Why the two axes moved. Orientation error is **monotone increasing in π**, not U-shaped as originally predicted: per-anchor standard deviation runs 3.9° at $\pi = 0.05$ to 47.5° at $\pi = 0.50$ (at $\Delta/\tau = 5$). Fewer crossers, but they sit further out along the normal, and the separation of class-conditional means along *n rises* as π falls — 2.17σ at $\pi = 0.05$ against 1.60σ at $\pi = 0.50$. At $\pi \to 0.5$ there is additionally no majority branch for LTS to commit to.

Since $\pi \leq 0.10$ means $d \geq 1.28\sigma$ and the minimum-mass rule caps $\pi \geq 0.05$ at $d \leq 1.64\sigma$, the orientation-useful anchors occupy a **shell**. The original distance axis sampled the wrong side of it.

> **A deployment consequence P must measure.** That shell is roughly 22% of the anchors that can detect anything at all, which are themselves 15–20% of randomly placed anchors — so on the order of **3–4% of anchors in deployment**. Pooling is weighted rather than thresholded, so higher-π anchors still contribute, but N_{95} under *random* placement may be far larger than N_{95} under placement by design. **Report both.**

9 Metrics

1. N_{50} and N_{95} per estimator per Δ, **under both by-design and random anchor placement**. [NEW] Headline.
2. Estimator ranking.
3. $N \mapsto$ orientation error and $N \mapsto |\hat{c} - c^{\star}|$. The first is partly previewed (§1); the second is untested.
4. False-positive rate under the permutation null.

Secondary: two parallel boundaries at separation $\in \{0.5, 2, 5\} \cdot r_{\text{hi}}$ — does mode-seeking resolve two modes or merge them? This sets the resolution limit for telling one gate from two. And a curved boundary (the deep-gate preimage): pool locally, then link, and report where the local-hyperplane approximation fails as a function of curvature $\times\, r_{\text{hi}}$.

10 The no-information guard

[NEW]

> Outside the operating region — $\Delta/\tau = 1.0$ at any π, or $\pi \gtrsim 0.35$ — the per-anchor rotation has standard deviation $\approx 52^\circ$, which is exactly what uniformly random directions give. Pooling uninformative *axial* estimates does not converge to the truth and does not converge to nothing: the mean direction of N random axes is itself uniformly random, so the pooled estimate lands on an arbitrary direction with an interval that *tightens* around it.

Pooling can therefore be confidently wrong in a way that per-anchor abstention never is. This is the second failure mode of that character in the method — the first being the $L < r_{\mathrm{hi}}$ degeneracy, where a ball spanning several routing cells reads as heteroscedastic noise and Stage 0 is blind to it.

P must include an explicit check that the anchor set carries information before it reports. The permutation null is the natural instrument, since a genuinely uninformative anchor set should be statistically indistinguishable from a permuted one. **Designing and calibrating that guard is part of P, not a follow-up.**

PART III

Pre-registered predictions

11 Stated before running; report hits and misses

#	Prediction	What falsifies it
S-1	The OOD detector classifies $> 80\%$ of our tangent-frame probes as *real* — trichotomy regime (i).	$< 80\%$; or probes are called OOD, which is regime (iii).
S-2	The scaffold's $\Delta/\sigma_{\mathrm{resid}}$ exceeds 2.5, so detection power is not the binding constraint.	A measured ratio below ≈ 1.5, which puts the real target under the floor.
S-3	Retention ρ at boundary-crossing anchors falls below $\rho_{\min}$, so the audit abstains where the signal is.	ρ comfortably above threshold — in which case the on-manifold tension is not binding here.
P-1	Estimator ranking $C > B > A$ at every Δ, with the $C - B$ gap largest at $\Delta = 1.5\sigma$.	Any reordering; in particular $B \geq C$ would mean thresholding costs nothing.
P-2	A real gate produces a **single mode** in the pooled offset c; resonant curvature produces a **comb** of modes spaced by its wavelength ℓ.	A comb for a gate, or a single mode for resonant curvature — either would put smooth rejection back on the per-anchor decision rules.
P-3	N_{95} under random anchor placement exceeds N_{95} under by-design placement by roughly the inverse shell fraction ($\sim$ 20–30$\times$).	A much smaller penalty, meaning higher-π anchors contribute more than the per-anchor numbers imply.
P-4	Orientation error scales like $\sqrt{d/m}$ in *intrinsic* dimension, so it degrades at $d = 4, 8$. [OPEN] (not run at $d = 2$)	Error flat in d, which would extend E1's dimension-independence result from ambient to intrinsic dimension.

PART IV

Outcomes — Experiment S, run 13 August

12 Results against the predictions

S5, S1, S3 and S4 were run on all four targets: our geospatial housing scaffold and the three published Slack datasets, using Slack's own `adversarial_models`. 150 anchors per target, 400 probes per anchor, frame and radius supplied by local PCA on the 60 nearest real points.

Target	**S5** audit	**S5** LIME	**S3** ρ	straddle rate	response values
COMPAS	0.980	0.000	0.997	0.013	1
German	0.900	0.000	0.987	0.253	2
Housing	0.860	0.000	0.996	0.473	400
Communities & Crime	**0.001**	0.000	0.985	0.007	2

S4 (detector leakage on real points) is 0.000 on every target, and **no anchor on any target** had ρ below $\rho_{\min}$.

	Prediction	Outcome
S-1	detector calls $> 80\%$ of our probes real	**HIT on 3 of 4.** COMPAS 0.98, German 0.91, Housing 0.83 — against **0.000** for LIME's own perturbation on every target. **MISS on Communities & Crime** (0.001).
S-2	$\Delta/\sigma_{\text{resid}} > 2.5$, so detection power is not binding	**FALSIFIED.** Median 1.60 (COMPAS), 1.88 (German), 1.95 (Housing). The real scaffold sits *at* the dip's floor and inside the LRT's 0.42–0.54 band. Power is a live constraint; pooling is not optional.
S-3	retention falls below $\rho_{\min}$; the audit abstains on its own target	**FALSIFIED, cleanly.** $\rho = 0.985$–0.997 everywhere, zero anchors below threshold. The explicit falsification condition did *not* fire.

> **The trichotomy resolves to (ii) on housing.** As the probe radius grows, invisibility decays gently ($0.94 \to 0.83$) while the fraction of anchors whose ball straddles the OOD boundary rises sharply ($0.10 \to 0.53$). **Both conditions hold at once**: at full k-NN radius, 83% of probes are invisible to the detector and 46–53% of anchors reach the gate. COMPAS is regime (i) — invisible but never crossing. German is mixed, at 25% crossing.

13 The Communities & Crime miss

Not a radius or frame artifact: visibility stays at 0.011 with the dimension cap raised to 30, and at 2% of the k-NN radius. The mechanism appears to be a **lattice** effect. These detectors are

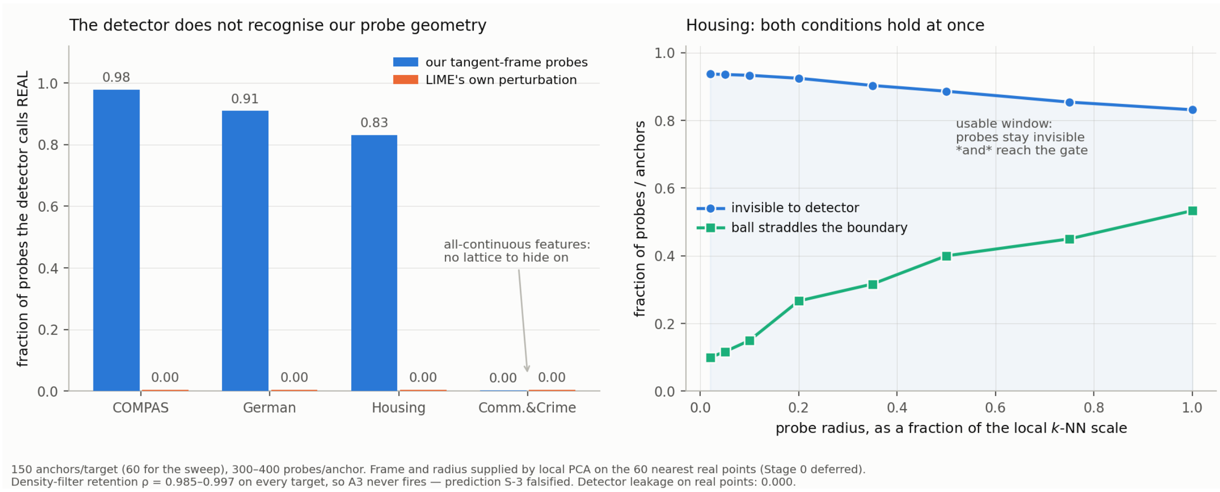

Figure 1: **Left:** the detector identifies 100% of LIME's own perturbations and misses 83–98% of ours. **Right:** on housing, invisibility and boundary-crossing hold simultaneously across the radius sweep.

extremely sensitive to displacement off the data's discrete structure — adding 10^{-3} of noise to a training point drops "called real" to 0.000/0.000/0.046, while genuinely held-out real points are still called real at ≈ 1.0. So the detector is not memorising; it keys on off-grid movement in quantised coordinates.

Tangent-frame probing largely respects that grid, because low-dimensional local PCA on mixed tabular data concentrates on the continuous directions. Measured off-lattice fraction: COMPAS 0.33, Housing 0.62, German 0.69, **Communities & Crime 0.96** — and its features are nearly all continuous, so there is no grid to stay on.

[OPEN] The ordering is right but this is not a clean single-factor explanation: German sits at 0.69 off-lattice and still reaches 0.91 visibility, so ambient dimension (101 against ≤ 29) is plausibly also at work. Directionally supported, not established.

14 Two consequences for the method

A discrete-response Stage 3 is a prerequisite for the Slack datasets. The published scaffolds expose **one or two distinct response values**; housing gives 400. Stage 3 tests the shape of a continuous residual distribution and does not apply to a binary response. This is the prerequisite flagged in the 31 July review, now confirmed rather than anticipated. Housing is unaffected.

A second, sharper claim is now available. Scaffolding attacks that key on LIME's ambient Gaussian perturbation are defeated by any probe family respecting the data's discrete structure — *but not on all-continuous data*. That is a statement about the attack's scope, and Communities & Crime is the counterexample that makes it non-trivial.

PART V

Outcomes — Experiment P core, run 13 August

15 A correction to this document's own null design

[NEW]

> **The permutation null cannot test existence.** It preserves the *multiset* of residuals, so the mixture fit — and with it the LRT statistic — is invariant to it by construction. Measured: real median 7.75 against permuted median 7.88. Estimator A therefore has zero power against it *by construction* rather than by weakness (0.000 at $\Delta/\tau = 5$), and testing A against it would be rigged.

Experiment P therefore runs **two** nulls. The **permutation** null is a null for *geometry* — does the response structure cohere with position across anchors. A matched **no-gate model** ($\Delta = 0$) on identical geometry is the null for *existence*, and all three estimators are comparable against it. Both are reported.

A second implementation correction: a Hough/grid peak is the wrong instrument at these anchor counts — 50 anchors over a 72×61 grid put roughly one anchor in the modal cell. Agreement is measured instead by a smooth grid-free pairwise kernel statistic $S = \sum_{ab} w_a w_b K_{\text{ang}} K_{\text{off}} / (\sum w)^2$, with offsets taken relative to the anchor centroid because an arbitrary origin amplifies direction error by the anchor's distance *along* the boundary.

16 P-1 — confirmed, including its specific clause

Power against the honest null, by-design placement, $\pi \leq 0.10$:

Δ/τ	N	A	B	**C**	**C−B gap**
1.0	100	0.112	0.188	**0.330**	+0.142
1.5	100	0.423	0.497	**0.973**	**+0.477**
2.5	25	1.000	1.000	1.000	0.000
5.0	5	1.000	1.000	1.000	0.000

The prediction was not merely $C > B > A$ but that the $C - B$ gap is *largest at* $\Delta = 1.5$ sd, where half the true detections fall below any per-anchor threshold. It is +0.477 there against +0.142 at the next-nearest cell. **Thresholding costs most exactly where the real target lives** — Experiment S measured the scaffold at 1.6–1.95 σ.

Against the *permutation* null the ranking holds everywhere: at $\Delta/\tau = 1.5$, $N = 100$, C = 0.907, B = 0.338, A = 0.027.

One reversal. At $\Delta/\tau = 2.5$, $N = 5$, A beats C (0.963 against 0.673): the LRT is already near-perfect per anchor at 2.5σ, so combining five is decisive while geometry needs more anchors to establish agreement. Everything saturates by $N = 25$, so this does not change the recommendation.

17 Recovery, and the no-information floor

Median orientation error of $\hat{n}$, by-design, $\pi \leq 0.10$, from $N = 5$ to $N = 100$:

Δ/τ	$N = 5$	10	25	50	100
1.0	41.6°	42.4°	38.9°	36.7°	34.3°
1.5	19.7°	19.1°	10.5°	8.1°	**6.3°**
2.5	4.5°	2.8°	1.8°	1.3°	**0.81°**
5.0	1.4°	0.9°	0.5°	0.4°	**0.28°**

Offsets converge in step: 0.052 at $\Delta/\tau = 1.5$ and 0.024 at 2.5 by $N = 100$ — that is 0.26σ and 0.12σ. At $\Delta/\tau = 1.0$ orientation does *not* converge, which is the no-information regime appearing in the recovery curve as well as in the power curve.

The permutation null is the no-information guard. At $\Delta/\tau = 1.0$, C's power against it is 0.10 at $N = 100$ — essentially nominal. When the anchors carry no coherent geometry, C does not beat permuted anchors. The guard the method needed is an instrument it already had.

18 P-3 — confirmed, and stronger than predicted

At $\Delta/\tau = 1.5$, $N = 100$: **by-design C** $= 0.96$**, random C** $= 0.00$. Not slower — dead. Estimator A over the same contrast: 0.70 against 0.74, unaffected. **Geometric pooling requires anchor placement; statistic-only pooling does not.**

The prediction was a 20–30× inflation in N_{95}. The reality at low Δ is qualitative rather than quantitative: anchors far from the boundary contribute noise directions that dilute agreement, while Fisher simply absorbs them as $p \approx 1$. At $\Delta/\tau = 2.5$ random placement still works but needs roughly an order of magnitude more anchors (N_{95}: 10 by design, 100 random).

This moves **boundary-seeking anchor placement** from a deferred deployment nicety to a prerequisite for the recovery claim. [OPEN]

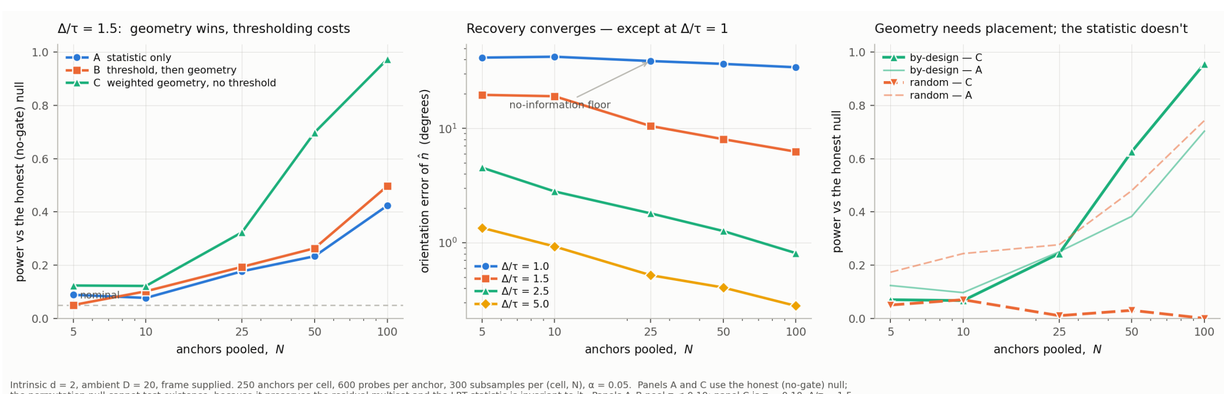

Figure 2: **Left:** estimator ranking at $\Delta/\tau = 1.5$. **Centre:** recovery against anchors, with the $\Delta/\tau = 1$ no-information floor. **Right:** by-design against random placement — geometry collapses, the statistic does not.

19 Two defects found and fixed in the harness

[NEW]

The first analysis showed orientation flat at $\sim 25°$; that was an averaging error across by-design and random placement in one pivot, not a result. And the shell placement rule collapsed all three π targets onto a single band, making the π axis degenerate; it now places at the exact distance realising each π with light jitter. Reported numbers are from the corrected run.

20 Order of execution

1. **S5, then S1** — done 13 August, all four targets; see Part III.
2. **S3** — done; it did not fire.
3. **Experiment P**, with the revised axes and the no-information guard.
4. S2 and S4; the COMPAS / German / Communities replication once the Slack repository is re-obtained.

Results are reported against the table in §13 — hits and misses, in that form.